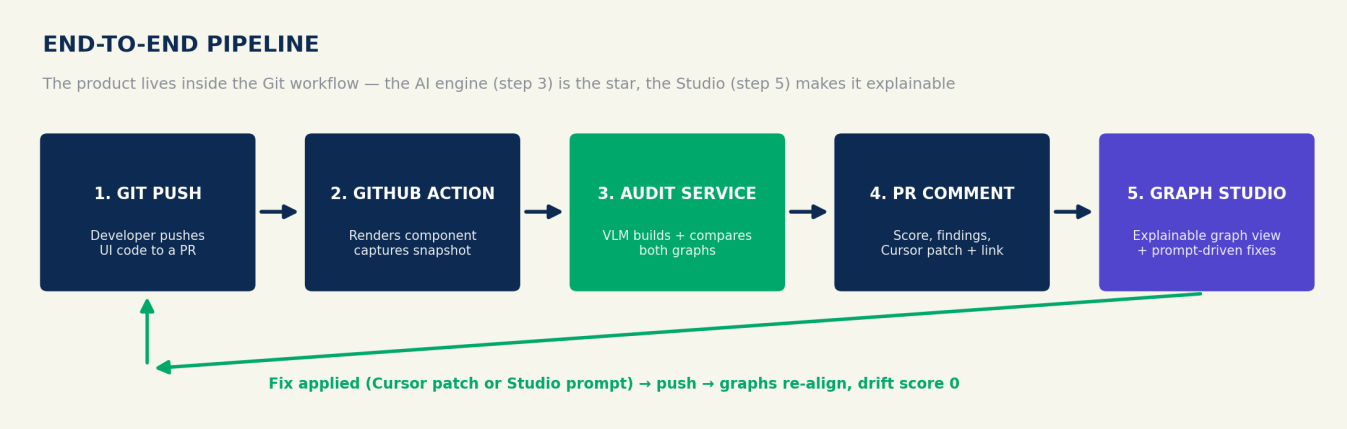
SynchronAIse

Execution Roadmap — Use Case 01: Automated Visual Auditing

RAISE Hackathon 2026 · Cursor Track · Submission deadline: **Sunday, July 5th, 12:00 PM**

1. What we are building

A **CI/CD-native design auditor** with a **Graph Reconciliation Studio** as its interface. The product lives inside the Git workflow: on every push, a GitHub Action renders the changed component, the AI engine builds and compares the two graphs (Figma intent vs code AST) and classifies every deviation, and the developer gets instant feedback — a PR comment with a Cursor-compatible patch, plus a link to the Studio where the drift is **visible, explainable and fixable via prompt**.



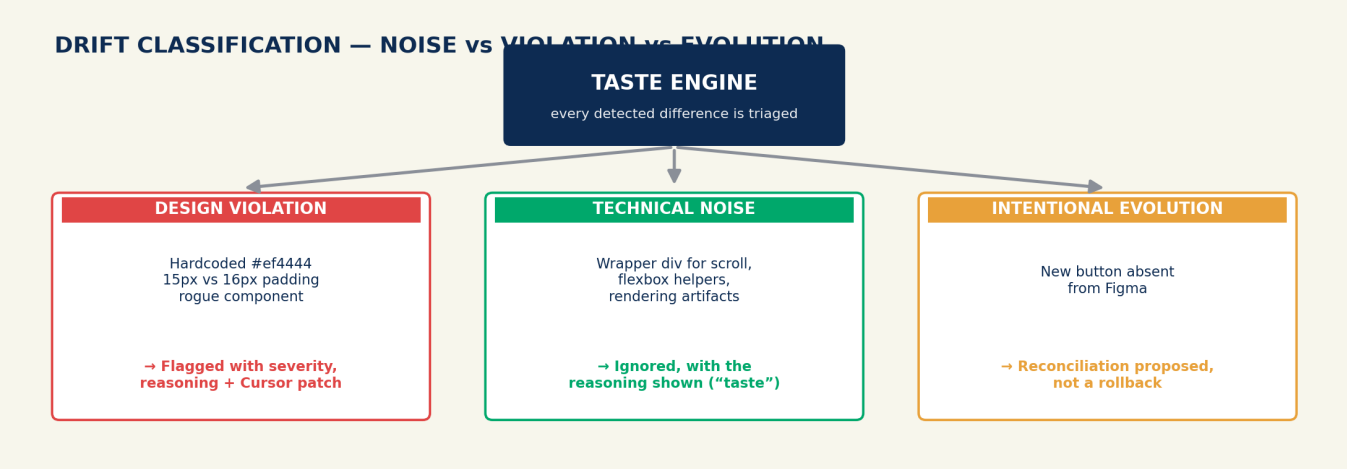
The full loop: push → snapshot → classification → feedback → fix in Cursor → green.

Strategic order — non-negotiable: the pipeline (steps 1–4) is the product and is built on Saturday. The Studio (step 5) is layered: graph view + node explainability first, prompt-driven fixes second. Any layer not demo-solid by Sunday 10:00 AM gets cut — the PR comment alone can still carry the demo.

Frozen scope. ONE Studio screen per audit — no PR list, navigation, auth, settings, multi-repo. Graph is read-only (no node dragging / re-mapping / graph editing — V2). Prompt fixes generate a patch, they never auto-commit. No QA Lens, no bidirectional sync, no dynamic Figma scraper, no CLI. **Why this is not a dashboard:** nothing here is passive metric display — every element is a step in an active workflow (see drift → understand reasoning → generate fix → verify green).

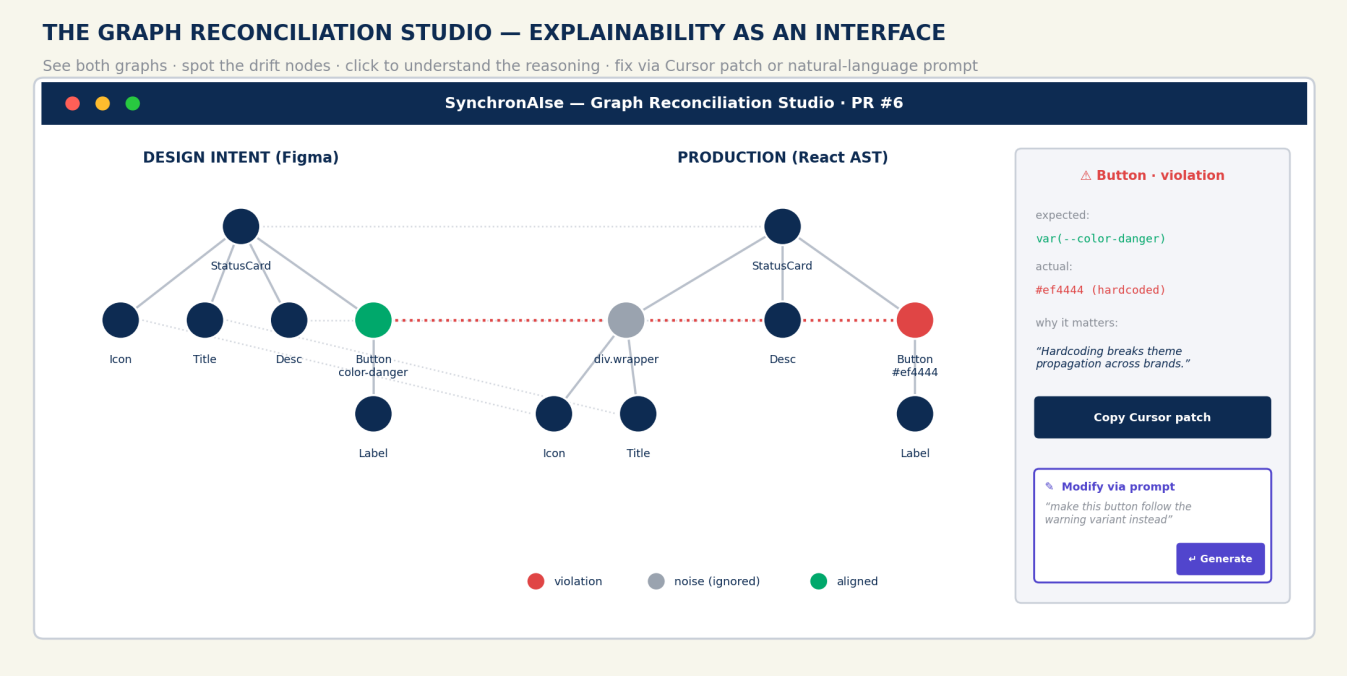
The Taste Engine — what makes us different

Every detected difference is triaged into three classes. Getting the middle one right — ignoring technical noise *with visible reasoning* — is the judged "taste" and the core of the project.



The Graph Reconciliation Studio — explainability as an interface

The frontend renders **both graphs side by side**: the Figma intent tree and the production code AST. Nodes are colored by classification (red = violation, grey = noise ignored, green = aligned); mismatch mappings are drawn between the two trees. Clicking a drift node opens the explanation panel: expected vs actual, the AI's reasoning, and the Cursor patch. A prompt box lets the user request a modification in natural language (“make this button follow the warning variant instead”) — the engine generates the corresponding patch.



Target Studio screen — built with react-flow (graph) + one explanation panel + one prompt box.

2. Team & responsibilities

Role	Owner	Main responsibility	Deliverable
R1 — AI Lead	AI PhD	Classification prompt: Noise vs Violation vs Evolution	Final prompt + validated JSON schema + measured accuracy
R2 — Data	Data expert	Golden dataset: tokens, hero component, drift PRs, bbox annotations	Hero component + 6 drift cases as real open PRs + ground truth
R3 — Backend	Big Data 1	Audit service: VLM orchestration, result storage, report data	/audit endpoint + stored results + GET /report/{id}
R4 — CI/CD	Big Data 2	The GitHub Action — critical path from hour one	Push → render + snapshot → call audit → post PR comment
R5 — Frontend	Platform eng	PR comment template (Sat) → Graph Studio: graph view, node panel, prompt box (Sun) + video + submission	Polished comment, one-screen Studio (react-flow), 1-min video, form submitted

Golden rule: every pair sharing an interface agrees on the JSON contract **before** writing code (§4). R3 mocks it in hour one so R4 and R5 never wait on R1.

3. Hero component & golden dataset (R2, top priority)

Component: an interactive StatusCard (icon + title + description + action button, with default / warning / danger variants). Rich enough for structural and token drift, fast to render in CI.

Hardcoded design system (~20 tokens): color-primary, color-danger, color-warning, color-surface, color-text · spacing-sm 8px, spacing-md 16px, spacing-lg 24px · font-heading, font-body · radius-md 8px…

#	PR / Case	Expected class	Why it's in the dataset
1	Hardcoded #ef4444 instead of var(--color-danger)	Design Violation	Obvious case, opens the demo
2	15px padding instead of 16px (spacing-md)	Design Violation	Pixel-level precision
3	Wrapper div added for scroll/flexbox handling	**Technical Noise (ignored)**	THE “taste” moment — a naive linter would flag it
4	Component rebuilt from scratch mimicking the StatusCard	Design Violation (rogue)	Subgraph matching
5	New secondary button absent from Figma	Intentional Evolution	Nuance: propose reconciliation, don't roll back
6	Cases 1+3 combined in the same file	Mixed	Proves the system separates signals

For each PR: drifted React code, CI-rendered screenshot, and **ground truth** (expected classification + precise violation + **bounding-box coordinates** of the drift zone, needed for the annotated screenshot). Real open PRs mean the demo repo tells the story by itself when judges browse it.

4. The JSON contract (freeze at H+1, everyone, 20 min max)

One payload drives everything: the PR comment, the report page, and the metrics script.

```
{
  "audit_id": "pr-6-run-3",  "pr_number": 6,  "drift_score": 72,
  "screenshot_url": "/artifacts/pr-6-run-3.png",
  "design_tree": { "id": "card", "children": [...] },   "code_tree": { "id": "card", "children": [...] },
  "findings": [{
      "type": "token_violation",
      "classification": "design_violation",
      "severity": "high",
      "location": "StatusCard.tsx:14",  "node_id": "btn-primary",
      "bbox": [120, 48, 340, 92],
      "expected": "var(--color-danger)",   "actual": "#ef4444",
      "reasoning": "Hex value matches token color-danger exactly; hardcoding breaks theme propagation.",
      "cursor_patch": {
        "prompt": "Replace the hardcoded #ef4444 on line 14 with var(--color-danger) from our design tokens.",
        "diff": "- color: #ef4444;  + color: var(--color-danger);" }
  }],
  "ignored_as_noise": [{
      "element": "div.scroll-wrapper",
      "reasoning": "Structural wrapper required for overflow handling; no visual or semantic impact." }],
  "evolution_proposals": []
}
```

• **design_tree / code_tree**: both graphs serialized with stable node ids — the Studio renders them directly; findings reference nodes via **node_id**, which drives the coloring.
• **bbox** kept as fallback for screenshot annotation if graph rendering slips.
• **cursor_patch** carries a paste-ready Cursor prompt *and* a suggested diff (rendered as a GitHub suggestion block).
• The backend **stores every audit** by audit_id; the report page is a pure renderer of this payload.
• Once frozen: R3 serves it hardcoded, R4 builds the Action against the mock, R5 designs comment + page against the mock, R1 makes the real LLM produce this format.

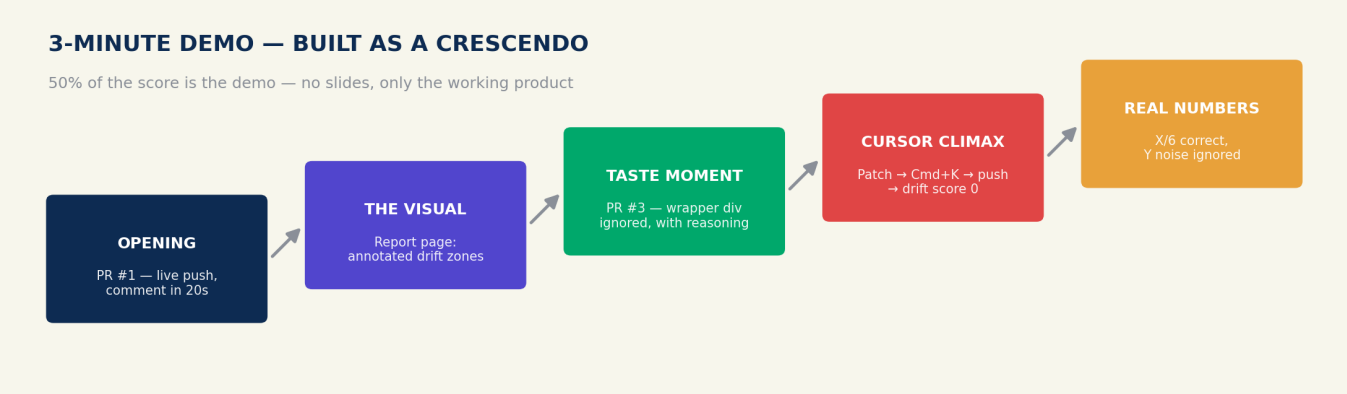
5. Timeline — Saturday 12PM → Sunday 12PM

Each role works in parallel against the mocked contract; red markers are the three hard checkpoints.

Slot	Focus	Key actions
Sat now → 3PM	Foundations	20-min kickoff: freeze JSON contract, create tool + demo repos. R4 ships a walking-skeleton Action (hardcoded comment) to prove trigger + permissions in hour one. R3 deploys mocked /audit. R5 designs the comment template. R1 sets up API access. R2 builds StatusCard + tokens.
Sat 3 → 7PM	Core loop	R4: Playwright render in CI → snapshot → /audit → formatted comment. R2 delivers PRs 1-3. R1 iterates the prompt until stable (3 identical runs). R3 wires the real VLM + storage. R5: comment on real payloads + Studio skeleton (react-flow renders the two mocked trees).
CHECKPOINT 7PM		**Pushing to PR #1 produces a real AI-generated comment, end to end.** If not, everyone converges on the blocker. The report page does NOT gate this checkpoint.

Slot	Focus	Key actions
Sat 7 → 10PM	Extend	R2 delivers PRs 4-6. R1 hardens across all 6 — case #3 is won or lost here. R4: audit only changed files, update comment on re-push (no spam). R5: Studio renders a real stored audit — trees colored by classification, click → explanation panel. Before leaving: push everything + written list of known bugs.
Sun 7 → 9AM	Hardening	E2E test of all 6 PRs, twice each; push-to-comment latency target < 60s. R1+R2 measure REAL metrics (the only numbers used in the pitch). R3: timeouts, retries, soft-fail comment. R5 (+R4): prompt box wired to the engine (node context + user prompt → generated patch) + green “resolved” state.
CUT 10AM		**Layered cut:** prompt box not solid → cut it (graphs + panel remain). Graph view not solid → cut the Studio, the PR comment carries the demo. No debate, no sunk-cost.
Sun 10 → 11AM	Demo & video	Rehearse the flow 3×: push → comment → Studio graphs → node click → prompt fix → push → green. R5 records the 1-min video of exactly that loop (YouTube/Loom). Plan B: full screen capture + one PR left pre-commented.
Sun 11 → 12PM	Submission	R5 submits the form (video, description, repos) — do NOT wait until 11:55. Everyone else: repo cleanup, no API keys in git history, .env.example, license, README “built during the event” section.
12:15 → judging	Pitch	3 minutes, two speakers max: 20s problem → 30s architecture → 90s live demo → 20s real metrics → 20s roadmap (QA Lens, bidirectional sync).

6. Demo script — 3 minutes, built as a crescendo



1. **Opening (PR #1):** push a commit live. “Twenty seconds later, SynchronAIse has reviewed the design.” → the comment appears: violation, reasoning, patch, report link.
2. **The visual (Studio):** open the Studio — two graphs side by side, one node burning red. Three seconds of silence — let it land. Click the node: the reasoning appears.
3. **The taste moment (PR #3):** “A classic linter would have flagged this wrapper div. Ours didn't — here's its reasoning.” → the *Ignored as technical noise* section.
4. **The prompt climax (PR #6):** type “fix this to use our danger token” in the Studio prompt box — the patch is generated; apply it via Cmd+K in Cursor, push → graphs re-align, everything green, **drift score 0**.
5. **Real numbers:** “On our test set: X/6 correct, Y noise cases correctly ignored.”

7. Risks & mitigations

Risk	Likelihood	Mitigation
LLM misclassifies case #3 (noise vs violation)	**High**	R1's main job; few-shot examples; fallback = whitelist of known structural patterns
Playwright-in-CI flaky	**High — critical path**	Pin browser versions; pre-built Docker image; fallback = pre-rendered screenshots committed per PR
Studio eats the schedule	Medium	Layered 10AM cut (prompt box first, then whole Studio); graph view is a pure renderer of the stored trees
Graph layout unreadable for deep trees	Medium	Hero component is shallow by design; react-flow auto-layout (dagre); collapse aligned subtrees
Prompt-fix returns a bad patch live	Medium	Rehearsed prompts only during the demo; patch is shown before applying, never auto-committed
Push-to-comment latency too long live	Medium	Compressed images, warm backend; one pre-commented PR as backup; recorded video
Non-JSON LLM output	Medium	response_format JSON if available, otherwise retry with the error in context
GitHub token permissions / rate limits	Medium	Walking skeleton proves permissions in hour one; dedicated bot token
Bboxes from the LLM are imprecise	Medium	Generous rounded highlights, not pixel-perfect boxes; ground-truth bboxes as demo fallback
API quota exhausted	Low	Two providers configured Saturday (Gemini + one fallback)
Running out of time	**Certain :)**	Sacrifice order: prompt box → Studio → PR #6 → PR #5 → comment-update-on-repush

8. Definition of Done — submission checklist

- ■ Public tool repo + public demo repo, README with architecture + run instructions + “built during the event” section

- [ ] All 6 drift cases live as open PRs in the demo repo, each with its AI-generated comment visible
- [ ] GitHub Action: push → snapshot → classification → PR comment, unassisted
- [ ] Studio renders the 6 demo audits: both graphs, node coloring, explanation panel (prompt box = bonus layer)
- [ ] Fix loop verified both ways: Cursor patch AND Studio prompt → apply → push → graphs green
- [ ] Real metrics measured and written in the README
- [ ] 1-min video uploaded + form submitted before 12:00 PM
- [ ] No API keys anywhere in git history
- [ ] Pitch rehearsed 3 times, under 3 minutes

SynchronAIse — pipeline first, Studio second, taste above all. Good luck at the 7PM checkpoint.